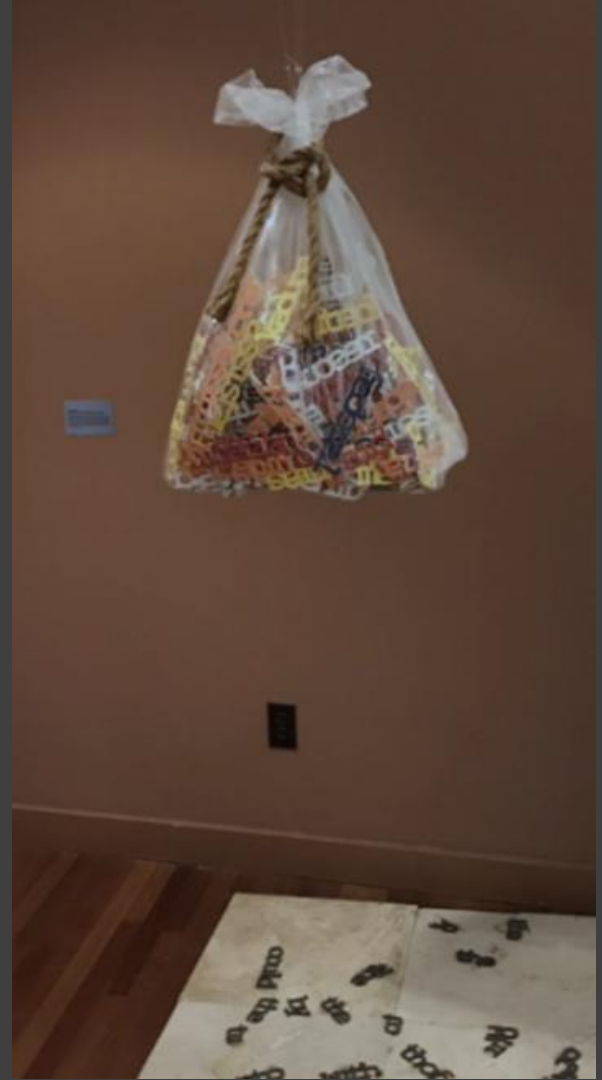
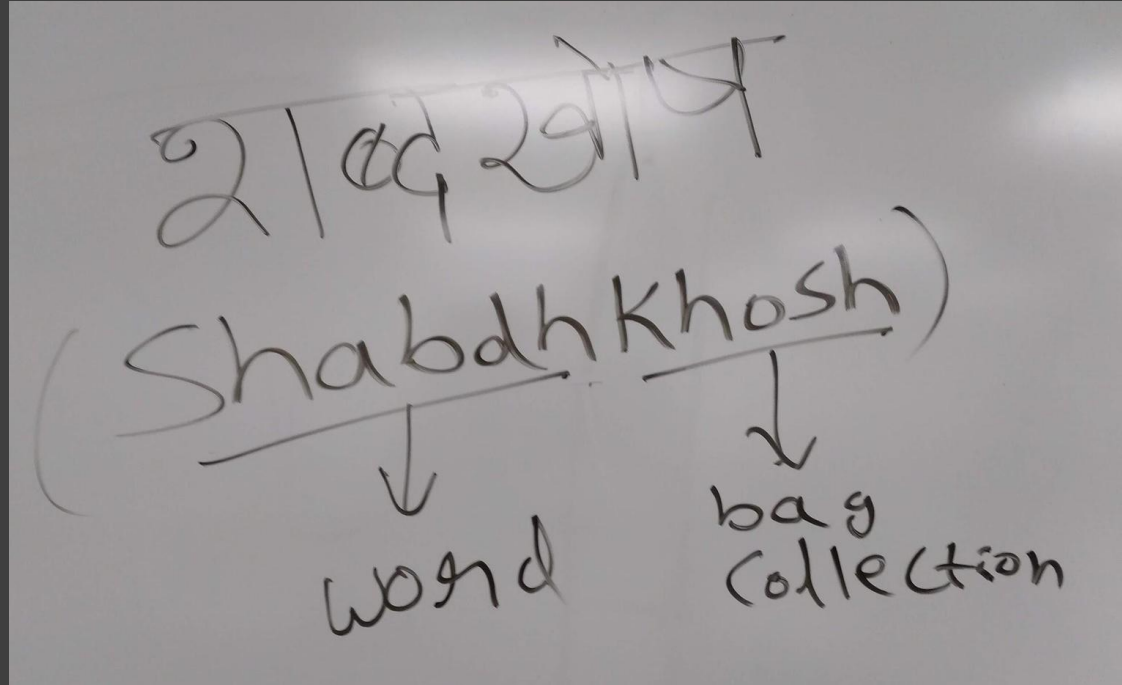




CS 1671 / CS 2071

Human Language Technologies

Session 6: Bag of words and n-grams

Michael Miller Yoder

September 10, 2026

GitHub Skills Workshop Series

featuring Pj Metz (GitHub Education)

- **GitHub 101**

Tuesday, September 15
3 p.m. - 5 p.m.

- **AI Literacy**

Wednesday, September 16
1 p.m. - 2 p.m.

- **GitHub Pages**

Thursday, September 17
12 p.m. - 1 p.m.

- **GitHub for Data Science**

Thursday, September 17
3p.m. - 4 p.m.



Education

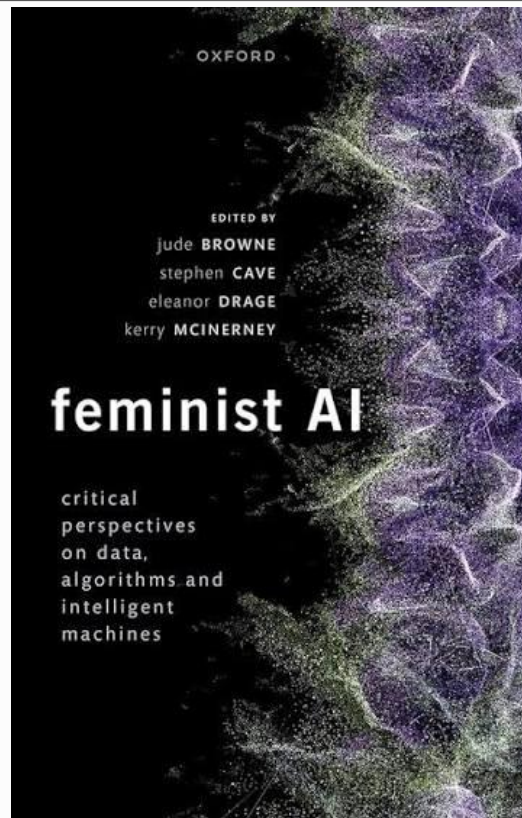


Learn more and register to attend at pi.tt/dsp-workshops



Digital Futures reading group: *Feminist AI*

- Hosted by the Sara Fine Institute here at SCI
- Selected essays in *Feminist AI* book:
<https://academic.oup.com/book/55103>
- Thursday 12-1pm meetings on Zoom
 - Sep 17, Oct 15, Nov 19
- Sign up [here](#)!



Assessments: quiz

- First in-class quiz is next class, **Tue Feb 15**
 - Covers readings from all the sessions up to that point
 - Looking over the reading is a great way to prepare
 - Session 3: J+M 2-2.6, 2.8, 2.10
 - Session 6: J+M 5.4, 11.1.1
 - Can cover content assigned in reading that is not discussed in class
 - Content from other sessions will not be included
- 3-4 questions
- Conceptual, not programming
- Lowest quiz score in the course will be dropped
- Quizzes are only 6% of your course grade total

Assessments: quiz

- In class on Canvas, 10 minutes to complete it (1-1:10pm)
- Allowed resources
 - Textbook
 - Your notes (on a computer or physical)
 - Course slides and website
- Resources not allowed
 - Generative AI
 - Internet searches
- If you won't be in class, let me know and I can accommodate

Assessments: project

- [Project idea form](#) to submit project ideas is **due next Tue Sep 15**
- Take a look at the example projects on the [project website](#). You can submit one or more of those for the form, or submit your own idea!
- Have a potential project idea that involves deriving insight from a dataset of text, or building an NLP system that can do something with text? You can submit it!
 - Ideas do not need to be well-formed
 - Ideas that have data already available are more realistic
- You will later choose from an **anonymized** list of project ideas on Project Match Day, Sep 22

Assessments: homework

- Homework 1 is **due next Thu Feb 17**
- Covers text processing and regular expressions in Python

Review activity

What do you remember about what machine learning is or what you can do with it?

Structure of this course

MODULE 1

Prerequisite skills for NLP

text normalization, linear alg., prob., machine learning

MODULE 2

Approaches

statistical machine learning

How text is represented

n-grams

NLP tasks

language modeling
text classification

MODULE 3

MODULE 4

MODULE 5

NLP applications and ethics

Structure of this course

MODULE 1	Prerequisite skills for NLP	text normalization, linear alg
	Approaches	How text is represented
MODULE 2	statistical machine learning	n-grams
MODULE 3		
MODULE 4		
MODULE 5	NLP applications and ethics	



Overview: Bag of words, n-grams

- Term-document matrices (bag of words)
- Cosine similarity

N-grams

Coding activity: clickbait n-gram document representations

Bag of words document representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

Documents Can Be Represented as Bags of Words

A BAG OF WORDS is a BAG or MULTISSET of the words in a document.

- Like a set, except that identical elements can appear multiple times
- You could also think of it like a `Counter` object in Python

```
bow = {'and': 23502,  
       'or': 12342',  
       'the': 54939508,  
       ...  
       'hippopotamus': 1}
```

- If you take out sequencing information, any document can be viewed as a bag of words.

Bags of Words Can Be Represented as Sparse Vectors

- So far, we've represented bags of words as dense MAPS, but sometimes it is useful to represent them as sparse vectors
- Let V be the set of all words in the document collection D . Then our BoW vectors for documents in D will have $|V|$ dimensions.
- Each dimension corresponds to a word **type** and the value at this dimension corresponds to the number of **TOKENS** of that type in the document that the vector is representing

Term-document matrices

Term-document matrix

		battle	good	fool	wit
	<i>As You Like It</i>	1	114	36	20
	<i>Twelfth Night</i>	0	80	58	15
	<i>Julius Caesar</i>	7	62	1	2
	<i>Henry V</i>	13	89	4	3

- Each cell is the count of term t in a document d ($\text{tf}_{t,d}$)
- Each document is a count vector in $\mathbb{N}^V$, a row above

Term-document matrix

		battle	good	fool	wit
	<i>As You Like It</i>	1	114	36	20
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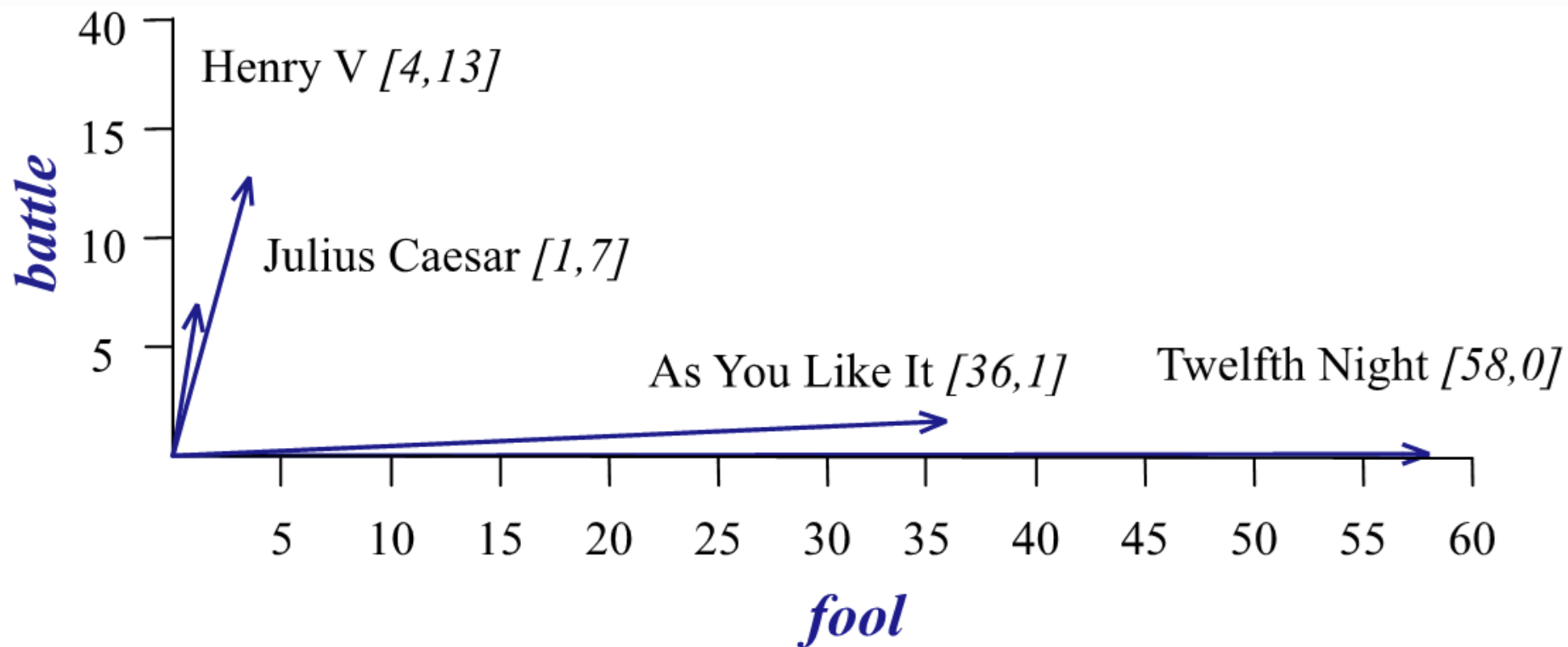
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- Each document is a **count vector** in $\mathbb{N}^V$, a row above

Term-document matrix

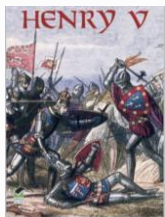
		battle	good	fool	wit
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	<i>Julius Caesar</i>	7	62	1	2
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- Documents are similar if their vectors are similar

Visualizing document vectors



Vectors are the basis of information retrieval

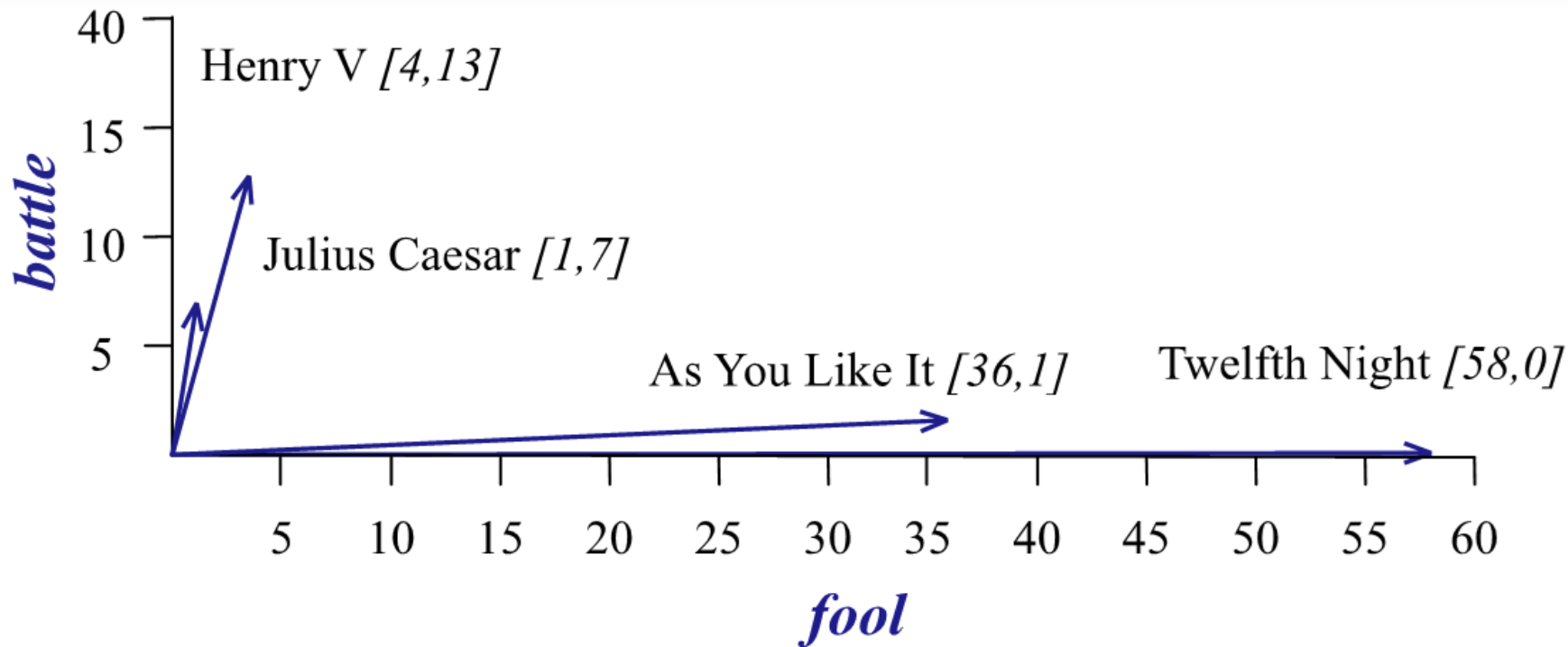


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- Vectors for comedies are different from tragedies
- Comedies have more *fools* and fewer *battles*

Cosine similarity

Measuring similarity between document vectors



Do we care about magnitude/word frequencies? (No)

Cosine measures the similarity between document vectors

Given 2 document vectors $\mathbf{v}$ and $\mathbf{w}$

- The **dot product** or **inner product** is usually used as the basis for similarity.

$$\mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^N v_i w_i = v_1 w_1 + v_2 w_2 + \cdots + v_N w_N$$

- $\mathbf{v} \cdot \mathbf{w}$ is high when two vectors have large values in the same dimensions.
- $\mathbf{v} \cdot \mathbf{w}$ is low (in fact 0) with zeros in complementary distribution.
- We also do not want the similarity to be sensitive to word-frequency.
- So normalize by vector length and use the cosine as the similarity

$$|\mathbf{v}| = \sqrt{\sum_{i=1}^N v_i^2}$$

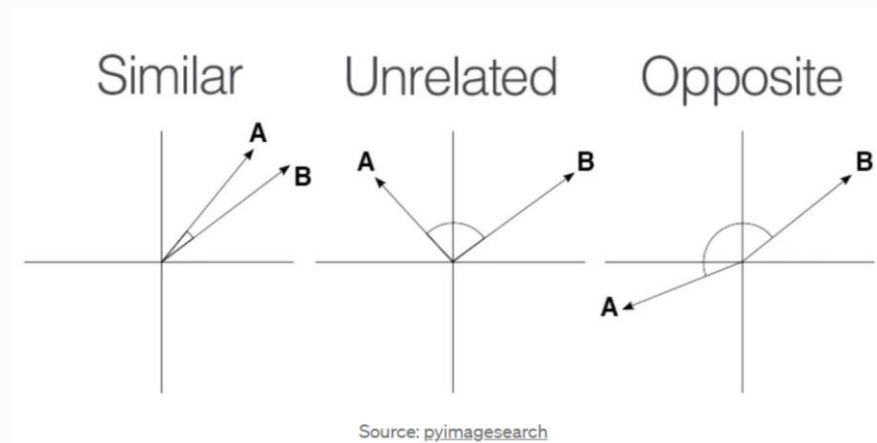
$$\frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}| |\mathbf{w}|} = \cos(\mathbf{v}, \mathbf{w})$$

Cosine as a similarity metric for vectors

-1: vectors point in opposite directions

+1: vectors point in same directions

0: vectors are orthogonal



But since raw frequency values are non-negative, the cosine for term-term matrix vectors ranges from 0–1

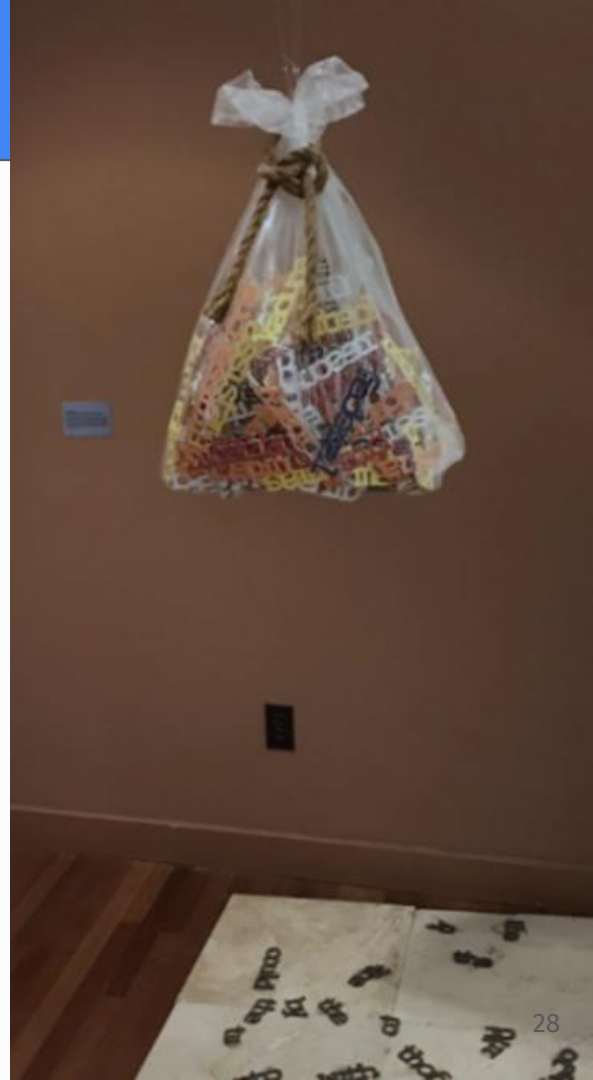
N-grams

N-grams

- A sequence of n words
- “Pittsburgh is cold in the winter.”
- Representations of documents:
 - Unigram: counts of all individual words
 - {Pittsburgh, is, cold, in, the, winter}
 - Bigram: counts of all sequences of 2 words
 - {Pittsburgh is, is cold, cold in, in the, the winter}
 - Trigram: counts of all sequences of 3 words
 - {Pittsburgh is cold, is cold in, cold in the, in the winter}
 - 4gram, etc
- Term-document matrix becomes even sparser!

Conclusion

- **Bag of words** representations of documents
 - Counts of terms in documents (no order info)
 - Can be represented in vector form as...
- **Term-document matrices**
- **Cosine** to measure similarity between vectors for documents or words
- **N-grams** are sequences of n words (tokens)



Coding activity:
clickbait n-gram document representations

N-gram document representations on JupyterHub

1. Go to this [nbgitpuller link](#) (also available on course website)
2. Log in with your Pitt username if necessary
3. Start a server with **TEACH - 6 CPUs - 45 GB**
4. Load custom environment at
`/ix1/cs1671-2026f/class-env/.pixi/envs/default`
5. This should pull the `cs1671_fall2026_jupyterhub` folder into your JupyterLab
6. Open `session6_clickbait_ngrams.ipynb`